

Odd numbers, square sums

Question 1 | Mathematical induction | 10 points

Prove the identity

For every integer $n \geq 1$, prove:

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

Prove by mathematical induction that $1 + 3 + 5 + \dots + (2n - 1) = n^2$ for every integer $n \geq 1$. State your base case and induction hypothesis. In the induction step, show how the sum for $k + 1$ follows from the sum for k , and justify the algebra. A numerical pattern or a diagram alone is not a complete proof.

What to include

1. A verified base case.
2. An induction hypothesis for an arbitrary integer $k \geq 1$.
3. The transition from k terms to $k + 1$ terms.
4. The algebra and final induction conclusion.

Write a connected proof

Explain why each step follows. You may use ordinary algebra and the definition of mathematical induction. State which part of your induction step uses the hypothesis.